Year 13 Further Maths - Probability

Similar questions - Counting arrangements and selections

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Seven red counters and three blue counters are arranged in a straight line in a random order. Find the probability that

(a) no two blue counters are next to each other, **[3]**

(b) every blue counter has at least two red counters between it and the nearest other blue counter. **[4]**

2 A set of 15 tiles contains 4 red tiles and 11 white tiles.

(a) Five tiles are selected at random without replacement. Find the probability that at least two are red. **[4]**

(b) All 15 tiles are arranged in a line in random order. Show that the probability that all the red tiles are next to each other is $\frac{4}{455}$. **[3]**

(c) Find the probability that the red tiles form exactly two separate pairs, each pair adjacent but the two pairs not touching. **[4]**

3 A bag contains 7 identical blue counters and 4 identical yellow counters.

(a) Three counters are selected at random without replacement. Find the probability that at least two are blue. **[3]**

All 11 counters are now arranged in a row in random order. Find the probability that

(b) all the yellow counters are next to each other, **[2]**

(c) no yellow counter is next to another yellow counter, **[3]**

(d) exactly three of the yellow counters are next to one another and the other yellow counter is not next to a yellow counter. **[3]**

(e) Explain whether the answer to part (d) would change if the yellow counters were numbered 1 to 4. **[1]**

Question 1

(a) Place the 7 red counters, then choose 3 of the 8 gaps for the blue ones: $\dfrac{\binom{8}{3}}{\binom{10}{3}} = \dfrac{56}{120} = \dfrac{7}{15}$

(b) Each of the two internal gaps between blue counters contains at least 2 red counters. Put 2 red in each of these two gaps; 3 red remain to be distributed over the 4 gaps: $\binom{6}{3} = 20$ ways. Probability $= \dfrac{20}{120} = \dfrac{1}{6}$

Question 2

(a) $1 - \dfrac{\binom{11}{5} + 4\binom{11}{4}}{\binom{15}{5}} = 1 - \dfrac{462 + 1320}{3003} = 0.407$

(b) Treat the red tiles as one block: $12! \times 4!$ arrangements out of $15!$: $\dfrac{4!}{15 \times 14 \times 13} = \dfrac{24}{2730} = \dfrac{4}{455}$

(c) Place the 11 white tiles; the red tiles form two pairs in two different gaps among the 12 gaps (including the ends): $\binom{12}{2} = 66$ ways. Probability $= \dfrac{66}{1365} = \dfrac{22}{455} = 0.0484$

Question 3

(a) $\dfrac{\binom{7}{2}\binom{4}{1} + \binom{7}{3}}{\binom{11}{3}} = \dfrac{84 + 35}{165} = \dfrac{119}{165} = 0.721$

(b) Treat the yellows as a block of 4 among 7 blue: 8 positions out of $\binom{11}{4} = 330$: $\dfrac{8}{330} = \dfrac{4}{165} = 0.0242$

(c) Choose 4 of the 8 gaps between blue counters: $\dfrac{\binom{8}{4}}{330} = \dfrac{70}{330} = \dfrac{7}{33} = 0.212$

(d) Block of 3 and a single yellow, in separate gaps among the 7 blue: $8 \times 7 = 56$ ways. Probability $= \dfrac{56}{330} = \dfrac{28}{165} = 0.170$

(e) No: both numerator and denominator are multiplied by 4!, so the probability is unchanged